

Summary Data on Acid Cropland and Lime Requirements in Uganda

GAIA

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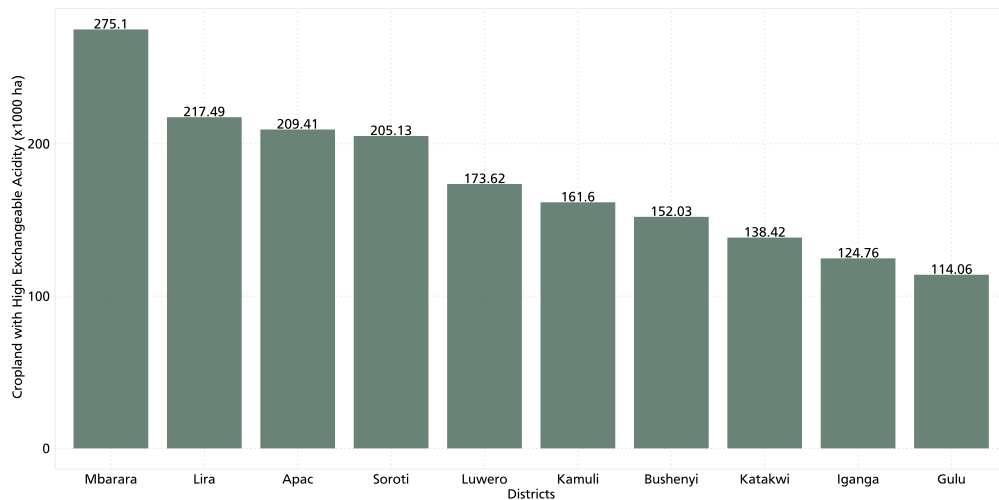
Outlook

- 1 Introduction
- 2 Acidic Cropland
- 3 Additional Production
- 4 Yield Loss
- 5 Yield Response
- 6 Lime Requirement

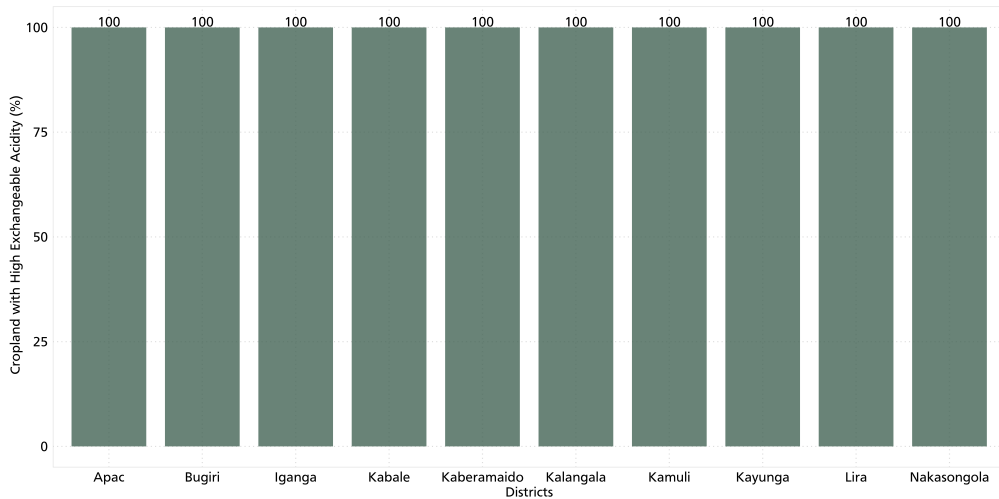
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Uganda**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production and production value from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

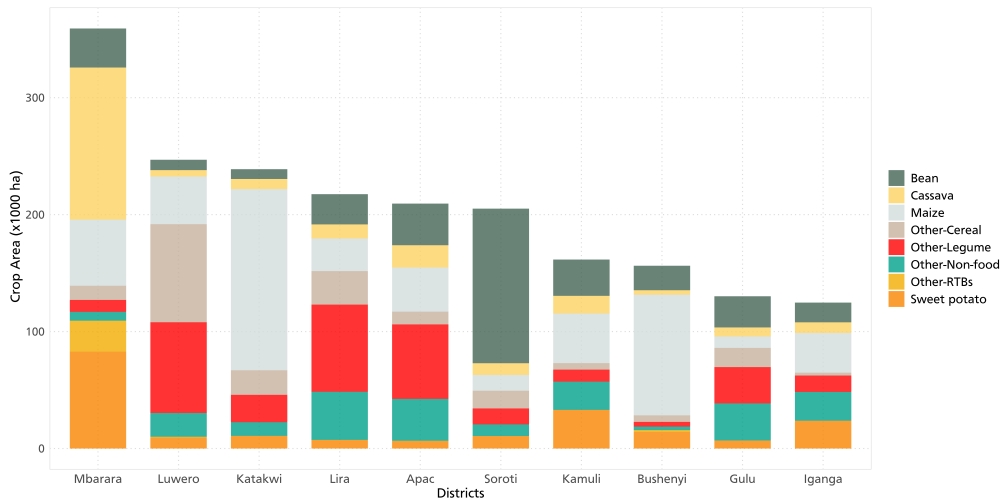
Top 10 Districts in Uganda with the highest acidic cropland



Top 10 Districts in Uganda with the highest acidic cropland (Percentage of total cropland)



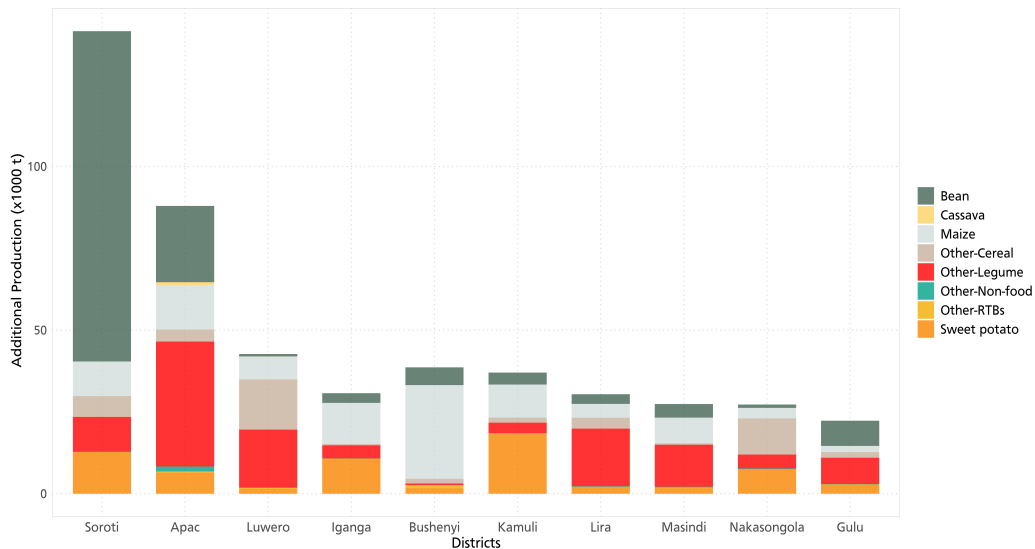
Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland



Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland (table)

Districts	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sweet potato
Apac	35.58	19.10	37.75	10.90	63.61	35.85	0.14	6.48
Bushenyi	20.96	3.83	103.15	5.63	3.99	3.03	1.55	14.17
Gulu	26.64	7.69	9.77	16.48	31.01	31.73	0.01	6.84
Iganga	16.87	9.12	33.92	2.44	14.07	24.55	-	23.80
Kamuli	31.03	15.13	42.46	5.51	10.28	24.15	0.01	33.02
Katakwi	8.35	8.69	154.86	21.04	23.40	11.69	-	10.80
Lira	25.88	11.97	27.96	28.65	74.52	41.15	0.06	7.30
Luwero	8.85	5.42	40.85	83.82	77.60	20.32	0.64	9.42
Mbarara	33.37	130.09	56.56	12.10	10.20	7.55	26.52	82.78
Soroti	132.17	10.06	13.46	15.17	13.57	9.97	-	10.72

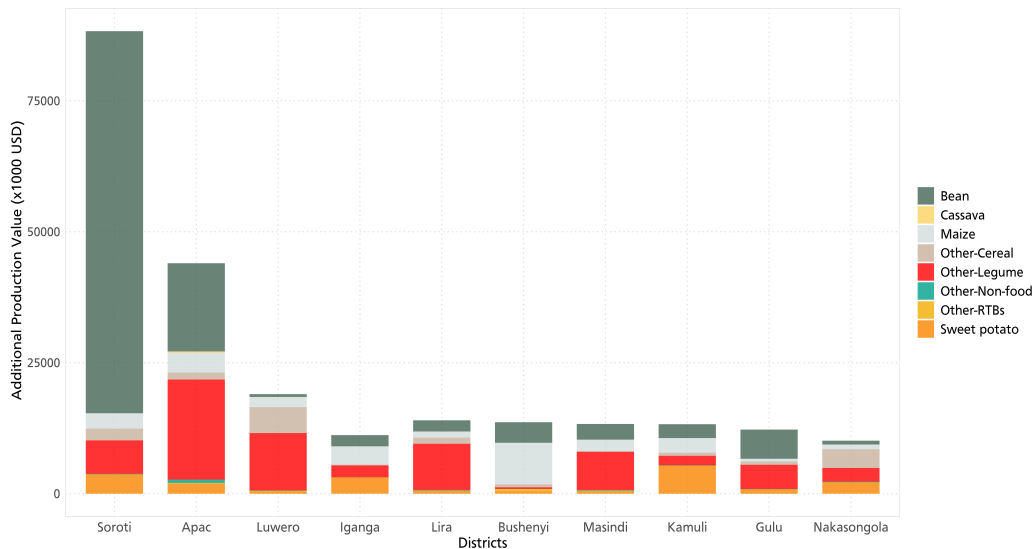
Potential for additional production (tone) in top 10 acidic Districts in Uganda



Potential for additional production (1000 tone) in top 10 acidic Districts in Uganda(Table)

Districts	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sweet potato
Apac	23.30	0.86	13.66	3.65	38.17	1.50	0.40	6.41
Bushenyi	5.41	-	28.64	1.43	0.49	0.01	0.95	1.67
Gulu	7.70	0.01	1.88	1.74	8.00	0.10	0.01	2.87
Iganga	2.95	-	12.59	0.38	4.01	0.03	-	10.76
Kamuli	3.65	-	10.11	1.54	3.15	0.10	-	18.45
Lira	2.96	-	4.23	3.32	17.59	0.19	0.01	2.10
Luwero	0.75	0.01	7.01	15.34	17.66	0.14	0.30	1.49
Masindi	4.14	0.02	7.86	0.40	12.84	0.24	0.24	1.66
Nakasongola	1.01	0.03	3.19	11.04	4.25	0.20	0.02	7.50
Soroti	100.96	0.03	10.56	6.35	10.60	0.14	-	12.73

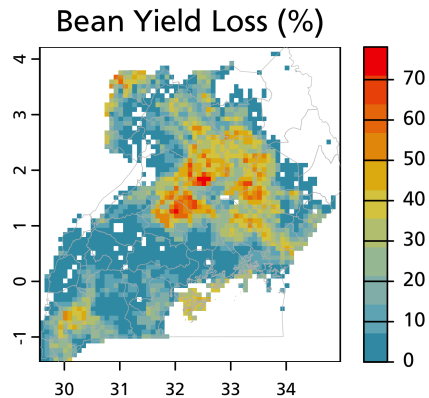
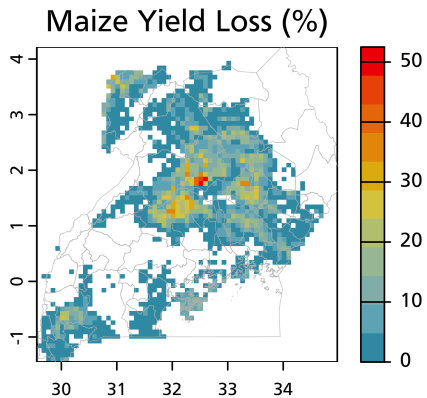
Potential for additional production value (US\$) in top 10 acidic Districts in Uganda



Potential for additional production value (1000 US\$) in top 10 acidic Districts in Uganda(Table)

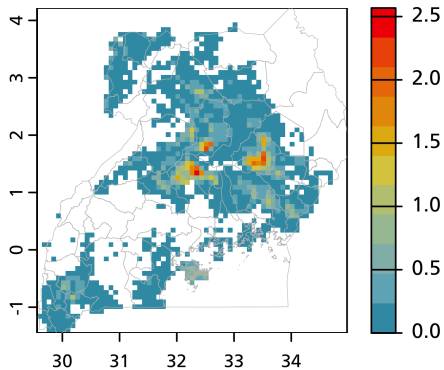
Districts	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sweet potato
Apac	16831.4	229.0	3793.4	1337.9	19099.0	660.0	173.1	1862.3
Bushenyi	3906.9	-	7957.4	551.2	309.2	21.0	407.9	486.2
Gulu	5565.1	1.3	523.5	621.7	4621.3	67.8	2.3	834.7
Iganga	2134.0	-	3498.7	141.8	2257.8	18.1	-	3128.1
Kamuli	2637.3	0.1	2808.6	587.5	1794.4	68.0	1.8	5364.4
Lira	2138.4	1.0	1176.4	1176.4	8786.7	119.5	6.4	610.7
Luwero	540.3	2.6	1946.8	4944.6	10938.6	74.1	127.4	434.5
Masindi	2990.8	4.4	2182.1	140.1	7269.3	139.2	101.0	481.2
Nakasongola	731.1	8.6	886.8	3577.6	2614.4	111.7	7.6	2180.9
Soroti	72932.8	6.9	2932.8	2241.8	6381.3	95.6	-	3700.6

Yield loss due to soil acidity for Maize and Bean in Uganda

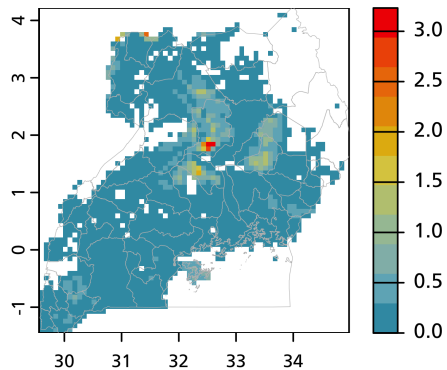


Yield response to lime application in Uganda

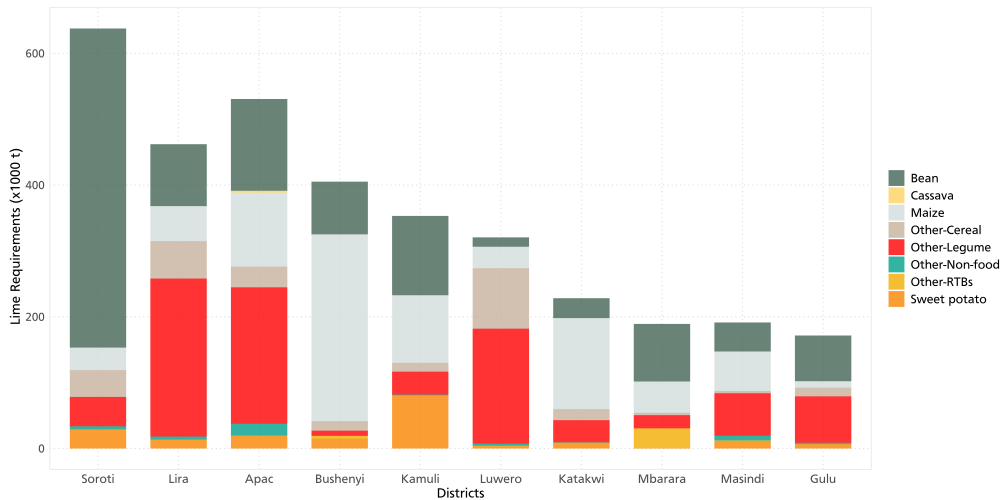
Maize Yield Response (t/ha)



Bean Yield Response (t/ha)



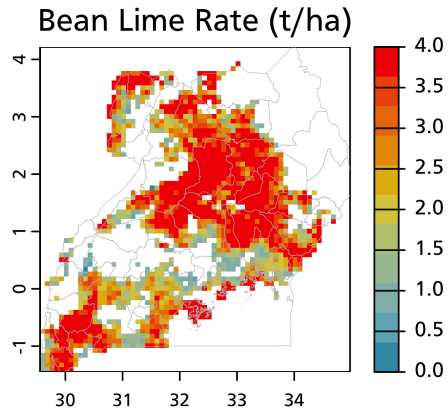
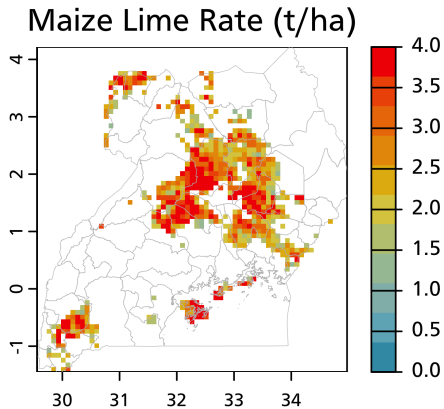
Lime requirements (t) for acid soil remediation in Uganda



Lime requirements (1000 t) for acid soil remediation in Uganda (table)

Districts	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Sweet potato
Apac	139.37	2.87	112.23	31.51	207.06	18.11	0.44	19.13
Bushenyi	80.04	-	283.59	14.54	7.61	0.37	4.37	14.74
Gulu	69.49	-	9.81	13.07	70.44	2.13	0.01	6.70
Kamuli	120.48	-	102.57	13.43	34.84	1.04	-	80.80
Katakwi	30.22	-	138.28	16.67	33.62	0.53	-	8.92
Lira	93.93	-	53.15	56.89	239.84	4.60	0.12	13.51
Luwero	14.28	-	32.53	91.84	174.62	3.23	0.83	3.33
Masindi	44.00	-	59.90	3.54	64.50	7.21	0.91	11.29
Mbarara	87.40	-	47.45	3.64	20.03	-	30.14	0.51
Soroti	484.51	-	34.27	40.66	44.66	4.57	-	29.13

Lime rate for Maize and Bean in Uganda



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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